Aditi Pore

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PROFESSIONAL SUMMARY

Third-year Electronics & Telecommunication Engineering student (Minor in AI & ML) at COEP Technological University, with hands-on experience developing and debugging embedded systems across microcontrollers (ESP32, ESP32-CAM), sensors, and analog signal-processing hardware. Skilled at integrating hardware, firmware, and APIs into working end-to-end systems, with analytical troubleshooting ability sharpened through a database internship and several independent embedded projects. Confident technical communicator, experienced explaining complex concepts to technical and non-technical audiences through leadership as Technical Secretary of the COEP Amateur Radio Club — seeking an Embedded Processing Field Applications Engineer internship at Texas Instruments to combine embedded hardware expertise with customer-facing technical problem-solving.

EDUCATION

B.Tech, Electronics & Telecommunication Engineering — Minor in AI & ML | *COEP Technological University, 2024–2028*

- JEE Mains — 96.98 percentile | MHT-CET — 99.65 percentile

EXPERIENCE

Database Intern — Softwin Infotech | *SQL & MySQL* | *6 Weeks, concluded July 2026*

- Diagnosed and resolved issues in real-world SQL/MySQL databases through structured troubleshooting and root-cause analysis, strengthening debugging and analytical thinking skills.
- Applied attention to detail and clear technical communication while collaborating with the team, quickly learning how database systems operate in a production business environment.

LEADERSHIP & ACTIVITIES

Technical Secretary — COEP Amateur Radio Club (VU2COE) | *June 2026–Present*

- Coordinate technical club activities and events end-to-end, managing logistics and communication across a team of members.
- Represent the club in interactions with external organizations, communicating technical radio and communications concepts clearly to audiences with varying technical backgrounds — building stakeholder relationships along the way.

Volunteer — COEP Amateur Radio Club (VU2COE) | *2024–2026*

- Supported technical club operations and community outreach programs, engaging directly with participants and the public before taking on the Technical Secretary role.

NGO Volunteer — Azad Nayakawadi Foundation, Kolhapur

- Engaged with community members to support foundation initiatives, developing interpersonal and relationship-building skills through direct outreach.

Volunteer — Mindspark 2024, COEP Technological University

- Assisted with event coordination and participant interaction, contributing to smooth execution and a positive attendee experience.

TECHNICAL PROJECTS

AI Vision Assistant for Visually Impaired | *ESP32-CAM · Groq LLaMA API · Google TTS · I2S*

- Built an embedded system on ESP32-CAM to capture surroundings and generate real-time spoken scene descriptions for visually impaired users, integrating camera capture, REST API calls to Groq LLaMA, Base64 image encoding, and I2S audio output.
- Debugged the full hardware-software integration pipeline — camera, API communication, OLED status display, and low-light flash support — demonstrating embedded firmware development, system integration, and peripheral interfacing.

Seven-Stage Analog EMG Signal Processing Circuit | *TL071 Op-Amps · Analog Front-End · EMG*

- Designed and debugged a seven-stage analog front-end for surface EMG signals — instrumentation amplifier (gain 201), band-pass filtering, 50 Hz notch filter, and envelope detector — to support prosthetic motor control.
- Validated the circuit through iterative testing, achieving ~10,000 V/V gain with a clean 0–5V DC output, demonstrating signal-processing design and hardware validation skills.

KHN State-Variable Audio Filter | *Op-Amp Based · Active Filter · ANC*

- Designed and tested a Kerwin-Huelsman-Newcomb active filter at 1 kHz with simultaneous LP, BP, and HP outputs and independent Q/frequency control, evaluating it as a low-cost analog alternative to DSP-based ANC systems.

Accelerometer-Based Air Mouse Using Microcontroller | *ESP32 · MPU6050 · Bluetooth HID*

- Integrated an MPU6050 accelerometer with ESP32 firmware to translate hand motion into BLE HID cursor commands, debugging sensor interfacing, click-button debouncing, and scroll-mode logic for a battery-powered, field-usable build.

SKILLS

- **Programming Languages:** Python, C, SQL
- **Embedded Systems & Hardware:** ESP32, ESP32-CAM, Microcontrollers, Sensor Interfacing, OLED, Embedded Firmware Development
- **Electronics & Signal Processing:** Analog Circuit Design, Op-Amps, Signal Conditioning, Active Filters, I2S Amplifiers
- **Communication Protocols:** I2C, I2S, UART, Bluetooth
- **APIs & Databases:** Groq LLaMA API, Google TTS, REST APIs, SQL, MySQL
- **Professional Skills:** Hardware & Software Debugging, Troubleshooting, Root Cause Analysis, Testing & Validation, System Integration, Technical Communication, Presentation Skills, Cross-Functional Collaboration, Team Management, Stakeholder Interaction, Community Outreach, Problem Solving, Rapid Learning, Adaptability, Initiative